

Positions

- 2026 – 2029 **Lovett Instructor**
Rice University, Houston TX
- 2024 – 2026 **Morrey Visiting Assistant Professor**
University of California, Berkeley CA
- 2018 – 2024 **Graduate Student Instructor & Graduate Student Research Assistant**
University of Michigan, Ann Arbor MI
- 2014 – 2018 **Research Assistant**
Chebyshev Laboratory at St. Petersburg State University, Russia

Education

- 2018 – 2024 **Ph.D. in Mathematics at the University of Michigan**
Advisor: Prof. A. Snowden *Tensorial measures on omega-categorical structures*
- 2017 – 2019 **Graduate study in Mathematics at St. Petersburg State University**
Advisors: Prof. G. Panina *Cohomology ring of polygonal linkages moduli space*
Prof. S. Vostokov *Cohomology of formal modules over local fields*
Prof. N. Vavilov *Overgroups of exterior powers of GL_n*
- 2012 – 2017 **Undergraduate study in Mathematics at St. Petersburg State University**
Advisor: Prof. S. Vostokov *Cohomology of formal modules*

Research Interests

- **Oligomorphic Groups, Tensor Categories, and Model Theory:** representation theory, categorical and model-theoretic interplay, structure theory of measures, monoidal abelian envelopes, and classification problem.
- **Representation Stability and Equivariant Geometry:** geometry and structure theory of infinite dimensional algebraic varieties with large group actions, and equivariant noetherianity.
- **Compactifications of $\mathcal{M}_{0,n}$ and Gromov–Witten theory:** connections to stability, category theory, and string theory.

Publications and Preprints

- **Model theory and $\text{Rep}(GL_t)$: definable topoi, measures, and tensor categories**
In preparation.
- **Mapping properties for oligomorphic tensor categories** with N. Harman and A. Snowden
In preparation.
- **Smooth representation theory of oligomorphic groups**
In preparation.
- **Upper bounds for measures on distal classes** with A. Snowden
Submitted, <https://arxiv.org/abs/2407.19131>
- **Arboreal tensor categories** with N. Harman and A. Snowden
Selecta Math. (N.S.) 31 (2025), no. 58.
- **Overgroups of exterior powers of an elementary group. Normalizers** with R. Lubkov
Documenta Math. 29 (2024), no. 5.

- **Dual infinite wedge is GL_∞ -equivariantly noetherian**
Available at <https://arxiv.org/abs/2008.09531>
- **Overgroups of exterior powers of an elementary group. Levels** with R. Lubkov
Linear and Multilinear Algebra, 72 (2024), no. 4.
- **Geometric presentation for the cohomology ring of polygon spaces** with G. Panina
St. Petersburg Math. J. 31 (2020), 59–67.
- **Explicit equations for the exterior square of the general linear group** with R. Lubkov
J. Math. Sci. (N.Y.), 243 (2019), no. 4, 583–594.
- **Compactifications of the space $\mathcal{M}_{0,n}$ associated with Alexander self-dual complexes: Chow rings, ψ -classes, and intersection numbers** with G. Panina
Proc. Steklov Inst. Math. 305, no. 1, 232–250.
- **Cohomology of formal modules over local fields** with S. Vostokov
Math. Notes 105 (2019), no. 1-2, 3–7.
- **Power filtration on morphisms of formal group law**
Lobachevskii J. Math. 39 (2018), no. 9, 1400–1402.
- **Alexander r -tuples and Bier complexes** with D. Jojić, G. Panina, and R. Živaljević
Publications de l’Institut Mathématique 104 (2018), Issue 118, 1–22.
- **Explicit constructions and arithmetic of local number fields** with S. Vostokov et. al.
Vestnik St. Petersburg Univ. Math. 50 (2017), no. 3, 242–264.
- **Cyclopermutohedron: geometry and topology** with A. Zhukova and G. Panina
Eur. J. Math. 2 (2016), no. 3, 835–852.
- **Lutz filtration as a Galois module** with S. Vostokov and R. Vostokova
Lobachevskii J. Math. 37 (2016), no. 2, 214–221.
- **The Lubin-Tate formal module in a cyclic unramified p -extension as a Galois module**
with S. Vostokov. J. Math. Sci. (N.Y.) 219 (2016), no. 3, 375–379.
- **The possibility of constructing a heat engine running on a cycle approaching the Carnot cycle** with M. Grinkrug. Int. J. Fund. Appl. Res. (RANH, Moscow), 12, pp 48-51.

--- Journals Refereed

Journal für die reine und angewandte Mathematik (Crelle), Selecta Mathematica.

--- Awards and Scholarships

- 2024 **The Cornwell Prize**, *doctoral student award*
The highest research prize of Math Department, the University of Michigan
- 2023 – 2024 **Rackham Predoctoral Fellowship**, *doctoral student award*
A prestigious award given by Rackham Graduate School at the University of Michigan
- 2019 – 2022 **Math Department Scholarship**, *doctoral student award*
University of Michigan, Department of Mathematics
- 2015, 2018 **Rokhlin Stipends for Young Mathematicians**, *graduate student award*
St. Petersburg Mathematical Society
- 2013 – 2017 **Scholarship of Chebyshev Laboratory**, *undergraduate student award*
St. Petersburg State University, Chebyshev Laboratory

--- Teaching Experience: Primary Instructor

- 2025 – 2026 **Introduction to Complex Analysis**, *Math 185*, lecturer, University of California, Berkeley
Two sections of upper-division undergraduate lecture-based course; designing the course, lecturing in sections of 40-50 students, and writing original homework assignments and exams.

- 2025, Fall **Honors Introduction to Abstract Algebra**, *Math H113*, lecturer, University of California, Berkeley
One section of upper-division undergraduate lecture-based course; designing the course, lecturing in a section of 41 students, and writing original in-depth challenging homework assignments and exams.
- 2024 – 2025 **Introduction to Abstract Algebra**, *Math 113*, lecturer, University of California, Berkeley
Three sections of upper-division undergraduate lecture-based course; designing the course, lecturing in sections of 40-50 students, and writing original homework assignments and exams.
- 2023, Aug. **Training for incoming Math instructors**, *Introduction to IBL*, facilitator, University of Michigan
Training and observing new instructors, leading lecture sessions, and designing *Inquiry-Based Learning*-friendly lesson plans.
- 2022, Fall **Calculus I**, *Math 115*, co-coordinator (1800+ students), University of Michigan
Mentoring and observing new instructors, redesigning the course, developing new class formats, and writing original *Inquiry-Based Learning*-type assignments and exams.
- 2022, Spring **Calculus II**, *Math 116*, coordinator and primary instructor (30+ students), University of Michigan
Coordinating the course and *Inquiry-Based Learning*-type teaching in a section of 20 students.
- 2018 – 2022 **Calculus I & II**, *Math 115 & 116*, primary instructor, University of Michigan
Five full semesters of *Inquiry-Based Learning*-type teaching in sections of 15–25 students
- 2017 – 2018 **Number Theory I & II, Class Field Theory**, *undergraduate extra courses*, primary instructor, Chebyshev Lab at St. Petersburg State University
Lecturing, facilitating problem solving sessions, course mini-projects advising, and writing homework problems.

Teaching Experience: Mentoring and Service

- 2024 – 2026 **Rep. Theory and Tensor Categories Seminar**, *organizer*, University of California, Berkeley
Co-organizing the seminar with Prof. V. Serganova.
- 2026, Spring **Advisor for Honors Undergraduate Theses**, *University of California, Berkeley*
Categorification and Combinatorics of Oligomorphic Groups by Ali Z Lone.
Identities on the Theta Ring of Simple Loopless Graphs by Aaron F. Burnham-Schmidt (UIUC).
- 2024 – 2025 **Reading Courses**, *University of California, Berkeley*
Mentoring 4 undergraduate students in reading courses on Representation Theory.
- 2020, Fall **Directed Reading Program**, *University of Michigan*
Matthew Wang, *Symmetries of Symmetric Groups*.
- 2020, May **School Math: Extra Practice**, *Khabarovsk Krai, Russia*
Math practice with high-schoolers over Zoom.
- 2019 – 2020 **Advisor for Master's Degree Thesis**, *St. Petersburg State University*
Nikita Elizarov (Universität Bielefeld) published his thesis *Lubin–Tate extensions and Carlitz modules*.
- 2017, Spring **Local Number Theory**, *Chebyshev Laboratory*
Organizer of a seminar for graduate students.
- 2013 **Math & Physics Summer Camp**, *Khabarovsk, Russia*
Course *Continuous methods in plane geometry* for middle and high-schoolers.

Funding and Research Groups

- 2018 – 2023 **Graduate Student Research Assistant Position**, *National Science Foundation, USA*
6 full semesters of GSRA positions at the University of Michigan
- 2017 – 2019 **Formal groups and arithmetic cohomology theories**, *Russian Science Foundation, 16-11-10200*

- 2016 – 2019 **Geometry and topology**, *Russian Science Foundation*, 16–11–10039
- 2016 **Formal modules and ramification theory**, *Russian Foundation for Basic Research*, 14–01–00393
- 2015 – 2016 **Mechanical linkages: Theory and Appl.**, *Russian Foundation for Basic Research*, 15–01–02021

Visits

- 2022, Dec. **Institute for Advanced Study**, *Princeton NJ*
Hosted by Prof. J. Draisma
- 2017, April **RIMS**, *Kyoto, Japan*
Hosted by Prof. A. Kirillov
- 2017, Spring **STEPS program**, *Tokyo University, Tokyo, Japan*
Hosted by Prof. Y. Mieda
- 2016, May **Mathematisches Forschungsinstitut Oberwolfach gGmbH**, *Oberwolfach, Germany*
Research in Pairs with D. Jojic, G. Panina, and R. Zivaljevic
- 2015, April **Bielefeld University**, *Bielefeld, Germany*
Hosted by Prof. A. Bak

Conference and Seminar Talks

- 2025 **Distal Classes Admit only Finitely Many Tensor Categories**, *Tensor Categories Seminar*, May 5, MIT
- 2024 **Oligomorphic Tensor Categories**, *Representation Theory and Tensor Categories Seminar*, September, UC Berkeley
Mini-course on recent developments in tensor categories and model theory.
- 2023 **Tensorial measures and ω -categorical theories**, *Logic seminar*, January 30, Wesleyan University & University of Connecticut
- **Arboreal Tensor Categories (poster)**, *Symmetric Tensor Categories and Representation Theory*, January 8–12, IPAM, Los Angeles
 - **Arboreal Tensor Categories**, *Algebra Seminar*, November 27, IU Bloomington
 - **Arboreal Tensor Categories**, *Representation Theory and Tensor Categories Seminar*, November 14, UC Berkeley
 - **Arboreal Tensor Categories**, *Algebra Seminar*, November 7, University of Oregon
 - **Arboreal Tensor Categories**, *Algebra Seminar*, October 2, University of Georgia, Athens
 - **Tensorial measures and ω -categorical theories**, *Logic Seminar*, September 20, University of Michigan
- 2018 **Symmetric–Exterior plethysms, vertex algebras and generalized Sato Grassmannians (poster)**, *Modern Algebraic Geometry*, July 21–25, Silkroad Mathematics Center, Beijing
- **Formal modules in constructible class field theory**, *Conference on Rings and Factorizations*, February 19–23, Graz, Austria
- 2017 **Cohomology of formal modules**, *Steklov Math. Institute*, November 28, Moscow, Russia
- **Explicit class field theory**, *Number Theory Seminar at HSE*, November 27, Moscow, Russia
 - **Formal modules in constructible class field theory**, *Mal'tsev Meeting at NSU*, November 20–23, Novosibirsk, Russia
 - **Intersection numbers of Chern classes of tautological line bundles on the moduli spaces of flexible polygons**, *St. Petersburg Topology Seminar*, October 18, POMI, St. Petersburg, Russia
 - **Classical plethysms and generalized grassmannians (poster)**, *Algebraic Modes of Representations The Canicular Days*, July 16–18, The Weizmann Institute of Science, Israel

- **Grassmannians and others homogeneous varieties in symmetric plethysms**, *Lie algebras, algebraic groups and Invariant theory*, January 31 – February 5, 2017, MSU and Independent University of Moscow, Moscow, Russia
- 2015 **Volume and lattice points counting for the cyclopermutohedron**, *The 5th German-Russian Week of the Young Researcher at the Moscow Institute of Physics and Technology in Dolgoprudnyj*, September 6-12, 2015, MIPT, Moscow, Russia
- **Volume and lattice points counting for the cyclopermutohedron**, *Polynomial Computer Algebra '2015*, April 13-18, 2015, Euler International Mathematical Institute, St.Petersburg, Russia
- **Lutz filtration as Galois module (poster)**, *Local arithmetic geometry in honor of Sergei V. Vostokov on the occasion of his 70th birthday*, May 18-22, 2015, Euler International Mathematical Institute, St.Petersburg, Russia

--- Conferences, Schools, and Workshops Attended

- 2021 **Chern–Simons and Other Topological Field Theories**, *MSRI*, Berkeley CA
- 2020 **Stability in Topology, Arithmetic, and Representation Theory**, *AMS Sectional Meeting*, online
- 2018 **Winter school on Local Geometric Langlands Theory**, *Paris 7 University*, France
- 2017 **Current Topics in the Theory of Algebraic Groups**, *Dijon University*, France
 - **Arbeitsgemeinschaft: Higher Gross–Zagier Formulas**, *Oberwolfach*, Germany
 - **Cluster Algebras in Mathematical Physics**, *Johannes Gutenberg University Mainz*, Germany
 - **Workshop on Quiver Grassmannians and their Applications**, *University of Wuppertal*, Germany
- 2013 **Summer School “Contemporary Mathematics”**, *Dubna*, Russia

--- Professional Development

- 2023 **UM/AMiIBL workshop**, *University of Michigan*
- 2020 **Intercultural Leadership Seminar**, *University of Michigan*